



SQL Data Analysis Case Study

1-Global World Layoff



by

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github

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Project Overview

This project analyzes global technology industry layoffs using SQL to uncover patterns and trends in workforce reductions over time.

The analysis follows an end-to-end data analysis process, starting with data cleaning and transformation, followed by exploratory data analysis (EDA) to identify trends across companies, industries, countries, and time periods.

Objectives

1.

Data
Cleaning

2.

EDA

3.

Insights



SQL Data Analysis Workflow

From Raw Data to Actionable Insights

Data Understanding

Understand the dataset structure, define analysis objectives, identify relevant variables, and establish key business questions

→ Step 1

Data Cleaning

Identify and handle duplicates, missing values, inconsistent formats, and invalid records to prepare reliable data for analysis.

→ Step 2

Data Transformation

Transform and organize cleaned data using SQL techniques such as CTEs, subqueries, joins, and aggregations.

→ Step 3

Exploratory Data Analysis

Explore trends, patterns, relationships, and anomalies using SQL queries and descriptive analysis.

→ Step 4

Insights & Findings

Interpret analytical results, identify key patterns, and translate findings into meaningful conclusions.

→ Step 5

Areas of Expertise

Introduction :

This project demonstrates an end-to-end SQL data analysis workflow, from preparing raw data to exploring trends and deriving meaningful insights. The analysis focuses on data quality, SQL manipulation, and exploratory analysis using a global tech layoffs dataset.

Data Cleaning

Identify and handle duplicates, missing values, inconsistent data, and formatting issues to prepare reliable data for analysis.

Advanced SQL

Apply window functions, ranking, rolling totals, and other analytical techniques to uncover deeper patterns in the data.

Query

Use CTEs, subqueries, joins, filtering, and aggregate functions to manipulate and analyze structured datasets.

EDA

Explore trends and patterns across companies, industries, countries, and time periods using SQL-based analysis.

Dataset Overview

Dataset that i use

This dataset contains 2,361 records of layoffs across companies in the global technology industry. It covers layoff events from March 2020 to March 2023, providing information on companies, locations, industries, workforce reductions, funding stages, and funds raised.

Dataset from:

<https://github.com/AlexTheAnalyst/MySQL-YouTube-Series/blob/main/layoffs.csv>



Dataset Size:

The raw dataset contains 2,361 layoff records across 9 variables before the data cleaning process.



Time Period:

The dataset covers layoff events recorded from March 11, 2020 to March 6, 2023.



Geographic Coverage:

The dataset captures layoffs across 60 countries and 191 locations, providing a global perspective on workforce reductions.



Key Variables:

Key variables include company, industry, location, total employees laid off, percentage laid off, funding stage, country, and funds raised.

Data Cleaning Roadmap



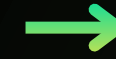
**Removed
Duplicates**



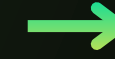
Standardize Data



**Handle Missing
Values**



**Remove Unnecessary
Columns**



Validate Clean Data

Data cleaning

```
with duplicate_cte as
(
select *,
row_number() over(
partition by company, industry, total_laid_off,
percentage_laid_off, `date`, stage, country, funds_raised_millions) as row_num
from layoff_staging
)
select * from duplicate_cte
where row_num >1 ;
```

- CTE (WITH) — creates a temporary result set for duplicate analysis.
- ROW_NUMBER() — assigns a sequential number to each record.
- PARTITION BY — groups records based on identical key attributes.
- WHERE row_num > 1 — identifies duplicate records.

country	funds_raised_millions	row_num
United States	339	2
United Kingdom	2000	2
Israel	45	2
Brazil	260	2
United States	6	2

Hapus data yang duplikat

- ## 1. Create new table for modification

```
CREATE TABLE `layoff_staging2` (  
  `company` text,  
  `location` text,  
  `industry` text,  
  `total_laid_off` double DEFAULT NULL,  
  `percentage_laid_off` text,  
  `date` text,  
  `stage` text,  
  `country` text,  
  `funds_raised_millions` int DEFAULT NULL,  
  `row_num` INT  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

- ## 2.Delete duplicate data in the new table

```
delete
from layoff_staging2
where row_num > 1;
```

- ## 2. Duplicate data already deleted

	company	location	industry	total_laid_off	percentage_laid_off	date	stage	country	funds_raised_millions	row_num



Data cleaning

2- Standardized Data

1. Delete excess space

```
select company, trim(company) from layoff_staging4;  
update layoff_staging4  
set company = trim(company);  
select * from layoff_staging4;
```

- TRIM() — removes leading and trailing whitespace.
- UPDATE — applies the cleaned values to the dataset.
- SELECT — verifies the standardized results.

2. Check Industry Column

```
select distinct industry from layoff_staging2  
order by 1;
```

industry
Consumer
Crypto
Crypto Currency
CryptoCurrency
Data
Education

Two records belong to the same industry but are labeled differently, so they need to be standardized for consistency.

3. Fix the industry column

```
update layoff_staging2  
set industry = 'Crypto'  
where industry like 'Crypto%';
```

Remove all the Crypto Currency and CryptoCurrency

Aerospace
Construction
Consumer
Crypto
Data

4. Check Location Column

```
select distinct location  
from layoff_staging2
```

location
Toronto
SF Bay Area
Dublin
Los Angeles
New York City
Sao Paulo
Washington D.C.

No problem in Location column

Data cleaning

2- Standardized Data

5. Check Country Column

```
select distinct country
from layoff_staging2
order by 1 desc;
```

United States.
United States

6. Fix the Country Column

```
select country,
trim(trailing '.' from country) as country1
from layoff_staging2
where country = 'United States.';
```

country	country1
United States.	United States
United States.	United States
United States.	United States
United States.	United States

7. Update table

```
update layoff_staging2
set country = trim(trailing '.' from country)
where country like 'United States%';
```

country
Vietnam
Uruguay
United States
United Kingdom
United Arab Emirates
Turkey

Country column save

8. Change Date Format

```
update layoff_staging2
set
`date` = str_to_date(`date`, '%m/%d/%Y');
```

date
2022-12-16
2022-07-25
2022-11-17
2023-01-27
2022-07-13

Change data type from date column

Data cleaning

2-Standardized Data

9. Change date format

```
alter table layoff_staging2
modify column `date` DATE;
```

ALTER TABLE — modifies the table structure.

MODIFY COLUMN — changes the column data type.

DATE — converts the date column into a proper date format.

Column: **date**

Definition:
date date

3-Handle Missing and Null

1. Check Industry Column

```
select *
from layoff_staging2
where (industry is null or industry = '');
```

company	location	industry
Airbnb	SF Bay Area	
Bally's Interactive	Providence	NULL
Carvana	Phoenix	
Juul	SF Bay Area	

There are 4 company that have missing industry

2. Make the missing value to null value

```
update layoff_staging2
set industry = NULL
where industry = '';
```

company	location	industry
Airbnb	SF Bay Area	NULL
Bally's Interactive	Providence	NULL
Carvana	Phoenix	NULL
Juul	SF Bay Area	NULL

To make the analysis easier



Data cleaning

3-Handle Missing and Null

3. Populate the data

```
select t1.company, t1.industry, t2.industry
from layoff_staging2 t1
join layoff_staging2 t2
    on t1.company = t2.company
where t1.industry is null
and t2.industry is not null;
```

- JOIN — matches records from the same company.
- Self-Join — compares records within the same table.
- IS NULL / IS NOT NULL — identifies missing values and available industry values.

company	industry	industry
Airbnb	NULL	Travel
Carvana	NULL	Transportation
Carvana	NULL	Transportation
Jul	NULL	Consumer

4.Update the data

```
update layoff_staging2 t1
join layoff_staging2 t2
    on t1.company = t2.company
set t1.industry = t2.industry
where t1.industry is null
and t2.industry is not null
```

company	industry	industry

No more missing value in industry column

5.Handle Missing value and null in total and percentage laid off column

```
select *
from layoff_staging2
where total_laid_off is null
and percentage_laid_off is null;
```

	total_laid_off	percentage_laid_off	date
	NULL	NULL	202
	NULL	NULL	202
	NULL	NULL	202
	NULL	NULL	202
	NULL	NULL	202



Data cleaning

3-Handle Missing and Null

6.Delete the null value

```
delete
from layoff_staging2
where total_laid_off is null
and percentage_laid_off is null;
```

company	location	industry	total_laid_off	percentage_laid_off	date	stage	country	funds_raised_millions	row_num

No more null value at this 2 column

4-Remove Unnecessary Column

is	row_num
1	
1	
1	
1	
1	
1	

This column dont have meaning

```
alter table layoff_staging2
drop column row_num;
```

Delete entire column

country	funds_raised_millions
United States	272
Ireland	35
Canada	21
United States	120
United States	242
United States	253



Data cleaning

3-Handle Missing and Null

6.Delete the null value

```
delete
from layoff_staging2
where total_laid_off is null
and percentage_laid_off is null;
```

company	location	industry	total_laid_off	percentage_laid_off	date	stage	country	funds_raised_millions	row_num

No more null value at this 2 column

4-Remove Unnecessary Column

is	row_num
1	
1	
1	
1	
1	
1	

This column dont have meaning

```
alter table layoff_staging2
drop column row_num;
```

Delete entire column

country	funds_raised_millions
United States	272
Ireland	35
Canada	21
United States	120
United States	242
United States	253

Final table

company	location	industry	total_laid_off	percentage_laid_off	date	stage	country	funds_raised_millions
Included Health	SF Bay Area	Healthcare	NULL	0.06	2022-07-25	Series E	United States	272
&Open	Dublin	Marketing	9	0.09	2022-11-17	Series A	Ireland	35
#Paid	Toronto	Marketing	19	0.17	2023-01-27	Series B	Canada	21
100 Thieves	Los Angeles	Consumer	12	NULL	2022-07-13	Series C	United States	120
10X Genomics	SF Bay Area	Healthcare	100	0.08	2022-08-04	Post-IPO	United States	242
1stdibs	New York City	Retail	70	0.17	2020-04-02	Series D	United States	253
2TM	Sao Paulo	Crypto	90	0.12	2022-06-01	Unknown	Brazil	250
2TM	Sao Paulo	Crypto	100	0.15	2022-09-01	Unknown	Brazil	250
2U	Washington D.C.	Education	NULL	0.2	2022-07-28	Post-IPO	United States	426
54gene	Washington D.C.	Healthcare	95	0.3	2022-08-29	Series B	United States	44
5B Solar	Sydney	Energy	NULL	0.25	2022-06-03	Series A	Australia	12
6sense	SF Bay Area	Sales	150	0.1	2022-10-12	Series E	United States	426
80 Acres Farms	Cincinnati	Food	NULL	0.1	2023-01-18	Unknown	United States	275
8x8	SF Bay Area	Support	155	0.07	2023-01-18	Post-IPO	United States	253
8x8	SF Bay Area	Support	200	0.09	2022-10-04	Post-IPO	United States	253
98point6	Seattle	Healthcare	NULL	0.1	2022-07-21	Series E	United States	247
99	Sao Paulo	Transportation	75	0.02	2022-09-20	Acquired	Brazil	244
Abra	SF Bay Area	Crypto	12	0.05	2022-06-30	Series C	United States	106
Absci	Vancouver	Healthcare	40	NULL	2022-08-09	Post-IPO	United States	237
Abiomed	San Francisco	Healthcare	72	0.15	2022-08-15	Post-IPO	United States	122
Abiomed	San Francisco	Healthcare	72	0.15	2022-08-15	Post-IPO	United States	122



EDA

1. Top 5 Companies with the Highest Number of Layoffs Over the Three-Year Period

```
select company, sum(total_laid_off) total_laid_off
from layoff_staging2
group by company
order by sum(total_laid_off) desc
LIMIT 5;
```

company	total_laid_off
Amazon	18150
Google	12000
Meta	11000
Salesforce	10090
Philips	10000

These are the five companies with the highest number of employee layoffs.

2. Which industry experienced the highest number of employee layoffs?

industry	total_laid_off
Consumer	45182
Retail	43613
Other	36289
Transportation	33748
Finance	28344

The Consumer sector had the highest number of employees laid off.

3. Which year experienced the highest number of employee layoffs?

years	total_laid_off
2022	160661
2023	125677
2020	80998
2021	15823

Layoffs peaked in 2022, with 160,661 employees laid off, followed by a significant decline in 2023.



EDA

4. Average Number of Employees Laid Off per company each Year

years	average_laid_off
2021	510
2023	420
2022	198
2020	170

2021 recorded the highest average layoffs per company, with 510 employees laid off.

5. Which country had the highest number of layoffs?

United States	256559
India	35993
Netherlands	17220
Sweden	11264
Brazil	10391

The United States recorded the highest number of layoffs, with 256,559 employees laid off



EDA

6. Ranking Companies by Total Layoffs per Year

2020

company	year	Total_laid_off	ranking
Uber	2020	7525	1
Booking.com	2020	4375	2
Groupon	2020	2800	3
Swiggy	2020	2250	4
Airbnb	2020	1900	5

Uber had the highest number of layoffs in 2020, with 7,525 employees affected.

2021

company	year	Total_laid_off	ranking
Bytedance	2021	3600	1
Katerra	2021	2434	2
Zillow	2021	2000	3
Instacart	2021	1877	4
WhiteHat Jr	2021	1800	5

Bytedance had the highest number of layoffs in 2021, with 3600 employees affected.

2022

company	year	Total_laid_off	ranking
Meta	2022	11000	1
Amazon	2022	10150	2
Cisco	2022	4100	3
Peloton	2022	4084	4
Carvana	2022	4000	5

Meta had the highest number of layoffs in 2022, with 11.000 employees affected.

2023

company	year	Total_laid_off	ranking
Google	2023	12000	1
Microsoft	2023	10000	2
Ericsson	2023	8500	3
Amazon	2023	8000	4
Salesforce	2023	8000	4
Dell	2023	6650	5

Google had the highest number of layoffs in 2023, with 12.000 employees affected.

EDA

7. Ranking Industries by Fund raised per Year

2020

company	year	funds_raised_millions	ranking
Uber	2020	98800	1
WeWork	2020	43500	2
Grab	2020	9900	3
OYO	2020	8000	4
Airbnb	2020	5400	5

2021

company	year	funds_raised_millions	ranking
Bytedance	2021	16800	1
Delivery Hero	2021	8300	2
Instacart	2021	2400	3
Dropbox	2021	1700	4
Katerra	2021	1600	5

2022

company	year	funds_raised_millions	ranking
Netflix	2022	487600	1
Meta	2022	26000	2
Uber	2022	24700	3
Tesla	2022	20200	4
Gopuff	2022	13600	5

2023

company	year	funds_raised_millions	ranking
Twitter	2023	25800	1
WeWork	2023	22200	2
Tencent	2023	12600	3
Rivian	2023	10700	4
Delivery Hero	2023	9900	5



Insight

1. Layoffs were highly concentrated in 2022–2023.

Layoffs increased substantially in 2022, reaching 160,661 employees, and remained high in 2023 with 125,677 employees laid off.

2. Geographic Concentration

The U.S. accounted for the majority of recorded layoffs, indicating that workforce reductions were geographically concentrated rather than evenly distributed across countries.

3. Industry Impact

Layoffs were not concentrated in a single technology-related segment, with Consumer, Retail, Transportation, and Finance all appearing among the most affected industries.

4. Annual Company Leaders

The company with the highest annual layoffs changed from year to year, indicating that large-scale workforce reductions were not concentrated in a single company throughout the period.



Thank You.

Thank you for reviewing my sql project. I'm always open to new opportunities and exciting collaborations.

Let's Connect:



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